What's new in DeployHub: Milestone 5

Live log streaming, metrics, health checks and rollback · 26 October 2026

> **In short:** logs now arrive live as they are written, the platform reports how busy its queue is and how fast deployments go, load balancers can ask whether an instance is ready for traffic, and a bad release can be undone with one click. Deployments themselves are still **simulated**: no code is cloned, built or run yet (that is Milestone 6).

1. Live logs (Server-Sent Events)

Before, the browser asked the server for new log lines every 1.5 seconds. Now it keeps one connection open and the server sends each line as soon as it is written.

- Every line carries its log id. If the connection drops, the browser reconnects by itself and continues from the last line it saw, with no gaps and no duplicates.
- If live streaming is not possible (for example, more than 5 streams open for one user), the page falls back to the old polling automatically, starting from the same line.
- The log panel shows which mode it is in: *live stream* or *polling*.
- The stream ends by itself once the deployment has finished and every line has been sent.

New endpoint: `GET /api/deployments/{id}/logs/stream`

2. Metrics

The backend now measures its deployment pipeline and publishes the numbers at `/actuator/prometheus`, in the format the monitoring tool Prometheus reads.

Metric	What it tells you
`deployhub_queue_jobs{state}`	Jobs waiting for a worker, waiting for a retry, running, or dead (gave up). Platform-wide, refreshed every 10 seconds.
`deployhub_queue_wait_seconds`	How long a job waited before a worker picked it up.
`deployhub_jobs_finished_total{outcome}`	Finished jobs by outcome. Their rate is the throughput.
`deployhub_jobs_retries_total`	Failed attempts that were scheduled for a retry.
`deployhub_deployments_duration_seconds`	Time from the deploy request to its final state.

- Only this machine can read the metrics by default, because they reveal load and internals. A request from the Mac's network address was tested and refused.
- The dashboard has a new **Last 60 minutes** panel: queue depth, average and longest wait for a worker, finished deployments, and success rate. The figures cover only *your own* deployments; only the queue depth is platform-wide, and it reveals nothing about other users.

New endpoint: `GET /api/dashboard/metrics`

3. Health checks

Milestone 7 puts DeployHub behind a load balancer, which needs two different questions answered:

Check	Question	What it looks at
`/actuator/health/liveness`	Is the process alive? (restart it if not)	Only the process itself. It deliberately ignores the database: restarting the app cannot fix a database outage.
`/actuator/health/readiness`	Should it receive traffic?	The database, and whether this process's worker is still reading the queue.

Both answer only *UP* or *DOWN*, with no internal details. Shutdown is now graceful: open log streams are closed first (browsers reconnect elsewhere), then running requests get up to 20 seconds to finish.

4. Rollback

A deployment that was live and has since been replaced now shows a **Roll back to this** button.

- A rollback creates a *new* deployment that redeploys the earlier one's commit. It goes through the same queue, duplicate-click protection, per-user limit, status history and audit log as a normal deploy.
- When it goes live it replaces the current live deployment, and it is shown as "Rollback to #N".
- Rolling back to the deployment that is already live, or to one that failed or was cancelled, is refused (HTTP 409).

New endpoint: `POST /api/deployments/{id}/rollback`

Tests

Suite	Before	Now
Backend on H2 (the quick, in-memory database)	113	**134**
Backend on real Oracle (Docker)	73	**91**
Frontend	77	**83**

One bug turned up along the way: the log-stream service stopped working after the application was stopped and started again, because it created its thread pool only once. The full test run exposed it; it is fixed and has its own test.

Checked end to end

Against a real running backend and frontend, with a throwaway database:

- Log lines arrived live, at 0.5, 3.6 and 6.6 seconds, as the deployment progressed.
- Deployed v1, then v2, then rolled back to v1: the rollback went live with commit v1.0.0 and v2 was marked as replaced.
- Sending the same rollback twice with the same key returned the same deployment; rolling back to the live deployment was refused.
- The new screens (metrics panel, rollback button) are covered by automated tests but were not looked at in a browser.

Still simulated or limited

- If a deployment was made without a commit, rolling back to it redeploys the branch's *current* code, not the exact old version. Milestone 6 will record exact commit IDs.

- A rollback re-runs the project's current build and start commands, because built packages are not stored yet.
- Counters and timers start from zero when the backend restarts; a Prometheus server adds them up over time (none is set up yet; planned for Milestone 7).

Also fixed just before Milestone 5

- Only the newest deployment of a project can go live. An older one that finishes late (for example after retries) is now skipped instead of replacing newer code.
- A replaced deployment keeps the time it originally went live.
- Pages keep refreshing through a short backend outage and recover by themselves.
- The log viewer stops on errors that won't go away (such as a deleted deployment) instead of retrying forever.
- The Deploy button also works on plain http, before HTTPS is set up.

Before you continue

- The next time the backend starts on your Oracle database, it adds one column and a few indexes (migration V4). Nothing existing is changed or removed.
- Milestone 6 (GitHub integration and real, isolated builds) is next. Setting up git first is recommended, so every change from then on can be compared and undone.

Full design reasoning: docs/adr/0005-observability-and-rollback.md · Endpoints and setup: README.md